Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 30)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

0 1 . 1 Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

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0 1 . 2 Explain the changes shown in Table 1.

[5 marks]

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0 1 . 3 Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

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0 1 . 4 Describe the early stages of a lithosere.

[1 mark]

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0 2 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 2 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 2 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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0 3 A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

0 3 . 1 Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

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0 3 . 2 Give **three** limitations of kick sampling.

[3 marks]

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0 3 . 3 Explain how a Surber sampler improves on kick sampling.

[2 marks]

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0 3 . 4 Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

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0 4 A conservation organisation is converting 700 ha of arable farmland on the coast into a wetland reserve for breeding and wintering birds (as at Wallasea Island).

0 4 . 1 Explain why habitat area and shape are important in the design of the reserve.

[4 marks]

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0 4 . 2 Explain how the reserve could be designed to provide a range of niches.

[3 marks]

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0 4 . 3 Explain why some species may need to be deliberately introduced while others colonise naturally.

[2 marks]

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0 4 . 4 Explain the value of a biological corridor linking the reserve to other wetlands.

[1 mark]

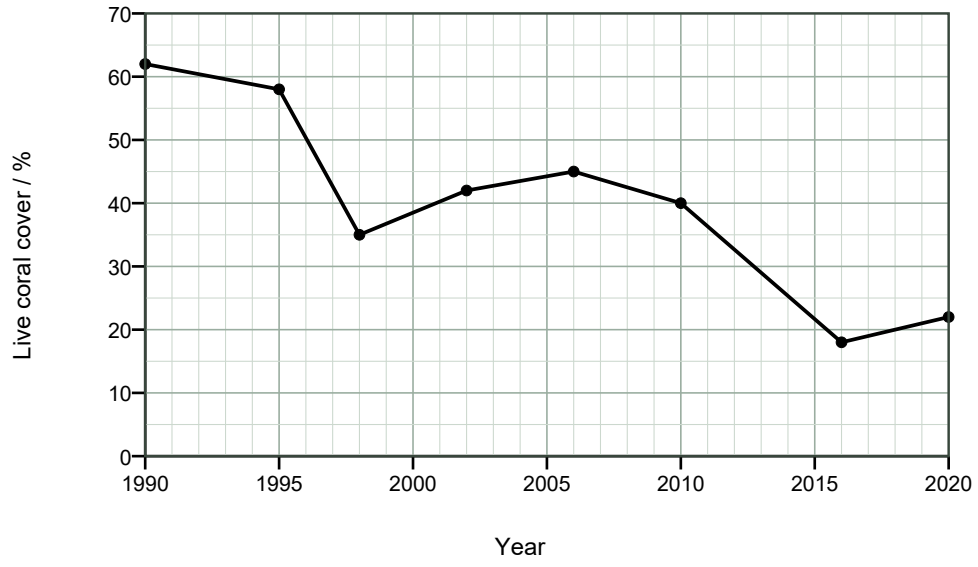
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0 5 Figure 1 shows the percentage of live coral cover on a reef near a growing coastal resort between 1990 and 2020, and the years of two major bleaching events.

Figure 1



0 5 . 1 Describe the changes in coral cover shown in Figure 1.

[2 marks]

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0 5 . 2 Explain what coral bleaching is and give **three** causes.

[4 marks]

0 5 . 3 Explain how the growing resort could threaten the reef other than through bleaching.

[3 marks]

0 5 . 4 Explain why a no-take zone on the reef could increase fish catches in surrounding areas.

[1 mark]

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0 6 . 1 Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

0 6 . 2 Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

0 6 . 3 Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

0 6 . 4 Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

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0 7 . 1 Describe **three** methods of reducing seabird by-catch in long-line fisheries.

[3 marks]

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0 7 . 2 Explain how restricting fishing effort and restricting fishing methods reduce overfishing.

[3 marks]

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0 7 . 3 Describe how captive rearing and release ('population seeding') can increase wild populations of lobsters or oysters.

[2 marks]

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0 7 . 4 Explain why collecting data on planktonic fish eggs and larvae helps to prevent population collapse.

[2 marks]

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0 8 Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 8 . 1 Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

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0 8 . 2 Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

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0 8 . 3 Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

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0 8 . 4 Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

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0 9 Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / $m^3\ ha^{-1}\ yr^{-1}$	4	16
Bird species	38	9
Ground flora species	60	4

0 9 . 1 Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

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0 9 . 2 Explain why commercial plantations are felled before the trees reach full size.

[2 marks]

0 9 . 3 Explain why there is little demand today for coppiced or pollarded wood.

[2 marks]

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1 0 . 1 Discuss the strategies that can be used to increase the sustainability of agriculture.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **permaculture**.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.

[25 marks]

OR

11 . 2 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS